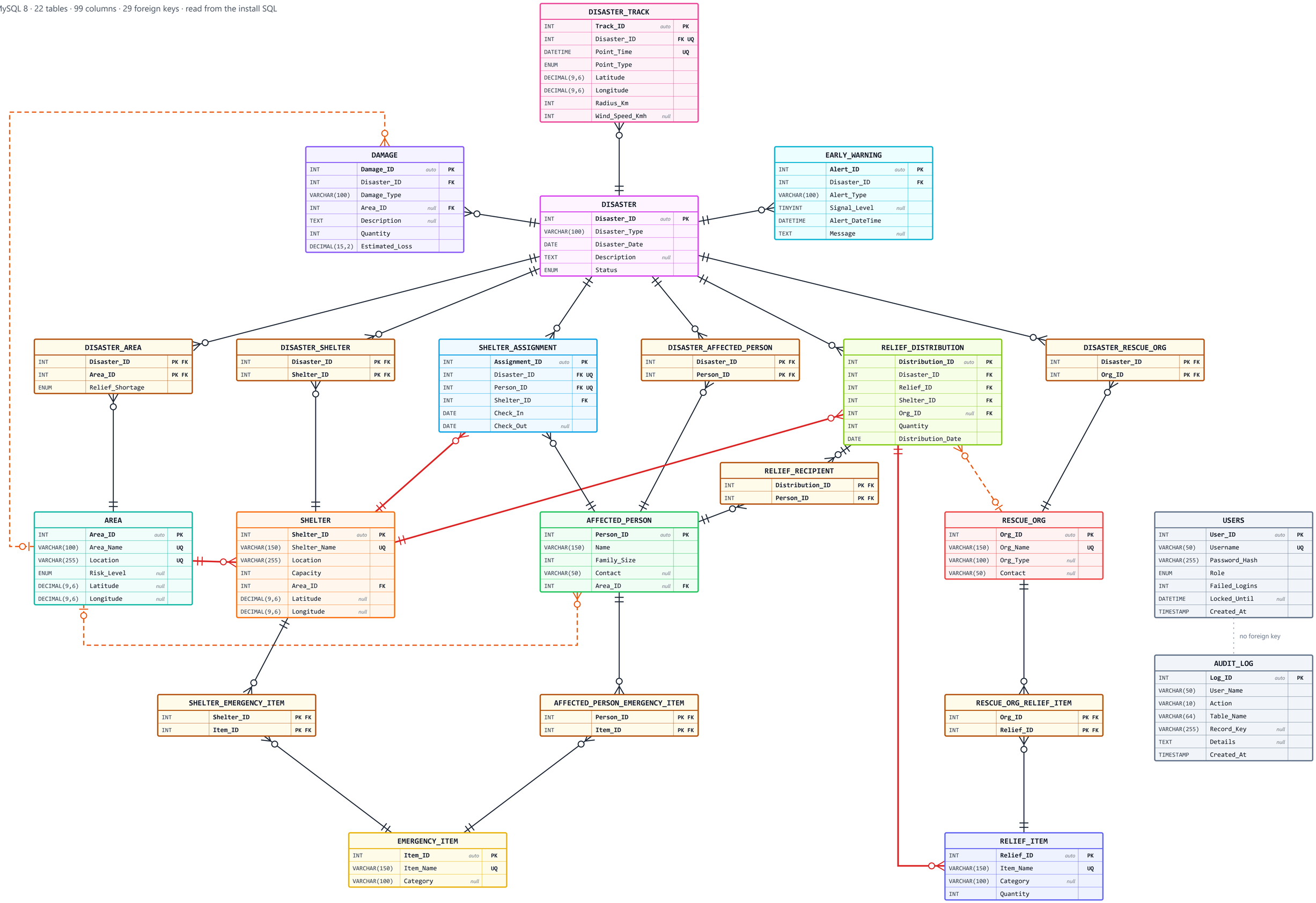
Disaster Mapping & Relief Management System — Entity-Relationship Diagram

MySQL 8 · 22 tables · 99 columns · 29 foreign keys · read from the install SQL



How to read this diagram

Every line is one FOREIGN KEY, drawn from the parent table (referenced) to the child table (holds the key). Boxes: **PK** primary key, **FK** foreign key, **UQ** part of a unique key, *null* column allows NULL, *auto* AUTO_INCREMENT.

Parent end: double bar = exactly one (key is NOT NULL) · circle + bar = zero or one (key allows NULL). Child end: circle + fork = zero or many.

— ON DELETE CASCADE (22): child rows are deleted too - - - SET NULL (3): the key is emptied ——— RESTRICT (4): deletion is refused

Colours: coloured = main tables · **amber = link tables** (their primary key is made of foreign keys) · **grey = stand-alone tables** (users, audit_log: no foreign keys).

Rules that need more than a foreign key (8 triggers, 16 CHECK constraints, 8 unique keys) are listed on page 2.

Specification behind the diagram

Read from disaster_and_relief_management.sql and inc/triggers.sql: 22 tables, 99 columns, 29 foreign keys, 8 unique keys, 16 CHECK constraints, 8 triggers.

Foreign keys (29)

#	Parent (referenced)	Child (holds the key)	Parent side	Child side	ON DELETE	ON UPDATE	Constraint
1	affected_person.Person_ID	affected_person_emergency_item.Person_ID	exactly one	zero or many	CASCADE	CASCADE	fk_person_item_person
2	affected_person.Person_ID	disaster_affected_person.Person_ID	exactly one	zero or many	CASCADE	CASCADE	fk_disaster_person_person
3	affected_person.Person_ID	relief_recipient.Person_ID	exactly one	zero or many	CASCADE	CASCADE	fk_recipient_person
4	affected_person.Person_ID	shelter_assignment.Person_ID	exactly one	zero or many	CASCADE	CASCADE	fk_assignment_person
5	area.Area_ID	affected_person.Area_ID	zero or one	zero or many	SET NULL	CASCADE	fk_person_area
6	area.Area_ID	damage.Area_ID	zero or one	zero or many	SET NULL	CASCADE	fk_damage_area
7	area.Area_ID	disaster_area.Area_ID	exactly one	zero or many	CASCADE	CASCADE	fk_disaster_area_area
8	area.Area_ID	shelter.Area_ID	exactly one	zero or many	RESTRICT	CASCADE	fk_shelter_area
9	disaster.Disaster_ID	damage.Disaster_ID	exactly one	zero or many	CASCADE	CASCADE	fk_damage_disaster
10	disaster.Disaster_ID	disaster_affected_person.Disaster_ID	exactly one	zero or many	CASCADE	CASCADE	fk_disaster_person_disaster
11	disaster.Disaster_ID	disaster_area.Disaster_ID	exactly one	zero or many	CASCADE	CASCADE	fk_disaster_area_disaster
12	disaster.Disaster_ID	disaster_rescue_org.Disaster_ID	exactly one	zero or many	CASCADE	CASCADE	fk_disaster_rescue_disaster
13	disaster.Disaster_ID	disaster_shelter.Disaster_ID	exactly one	zero or many	CASCADE	CASCADE	fk_disaster_shelter_disaster
14	disaster.Disaster_ID	disaster_track.Disaster_ID	exactly one	zero or many	CASCADE	CASCADE	fk_track_disaster
15	disaster.Disaster_ID	early_warning.Disaster_ID	exactly one	zero or many	CASCADE	CASCADE	fk_early_warning_disaster
16	disaster.Disaster_ID	relief_distribution.Disaster_ID	exactly one	zero or many	CASCADE	CASCADE	fk_distribution_disaster
17	disaster.Disaster_ID	shelter_assignment.Disaster_ID	exactly one	zero or many	CASCADE	CASCADE	fk_assignment_disaster
18	emergency_item.Item_ID	affected_person_emergency_item.Item_ID	exactly one	zero or many	CASCADE	CASCADE	fk_person_item_item
19	emergency_item.Item_ID	shelter_emergency_item.Item_ID	exactly one	zero or many	CASCADE	CASCADE	fk_shelter_item_item
20	relief_distribution.Distribution_ID	relief_recipient.Distribution_ID	exactly one	zero or many	CASCADE	CASCADE	fk_recipient_distribution
21	relief_item.Relief_ID	relief_distribution.Relief_ID	exactly one	zero or many	RESTRICT	CASCADE	fk_distribution_relief
22	relief_item.Relief_ID	rescue_org_relief_item.Relief_ID	exactly one	zero or many	CASCADE	CASCADE	fk_org_relief_item
23	rescue_org.Org_ID	disaster_rescue_org.Org_ID	exactly one	zero or many	CASCADE	CASCADE	fk_disaster_rescue_org
24	rescue_org.Org_ID	relief_distribution.Org_ID	zero or one	zero or many	SET NULL	CASCADE	fk_distribution_org
25	rescue_org.Org_ID	rescue_org_relief_item.Org_ID	exactly one	zero or many	CASCADE	CASCADE	fk_org_relief_org
26	shelter.Shelter_ID	disaster_shelter.Shelter_ID	exactly one	zero or many	CASCADE	CASCADE	fk_disaster_shelter_shelter
27	shelter.Shelter_ID	relief_distribution.Shelter_ID	exactly one	zero or many	RESTRICT	CASCADE	fk_distribution_shelter
28	shelter.Shelter_ID	shelter_assignment.Shelter_ID	exactly one	zero or many	RESTRICT	CASCADE	fk_assignment_shelter
29	shelter.Shelter_ID	shelter_emergency_item.Shelter_ID	exactly one	zero or many	CASCADE	CASCADE	fk_shelter_item_shelter

CHECK constraints (16)

Table	Constraint	Rule
affected_person	chk_person_family	Family_Size > 0
area	chk_area_lat	Latitude BETWEEN -90 AND 90
area	chk_area_lng	Longitude BETWEEN -180 AND 180
damage	chk_damage_qty	Quantity >= 0
damage	chk_damage_loss	Estimated_Loss >= 0
early_warning	chk_warning_signal	Signal_Level BETWEEN 1 AND 10
relief_item	chk_relief_qty	Quantity >= 0
shelter	chk_shelter_capacity	Capacity > 0
shelter	chk_shelter_lat	Latitude BETWEEN -90 AND 90
shelter	chk_shelter_lng	Longitude BETWEEN -180 AND 180
shelter_assignment	chk_assignment_dates	Check_Out IS NULL OR Check_Out >= Check_In
relief_distribution	chk_distribution_qty	Quantity > 0
disaster_track	chk_track_lat	Latitude BETWEEN -90 AND 90
disaster_track	chk_track_lng	Longitude BETWEEN -180 AND 180
disaster_track	chk_track_radius	Radius_Km > 0
disaster_track	chk_track_wind	Wind_Speed_Kmh >= 0

Tables (22)

Table	Kind	Cols	Primary key	Checks	Triggers
affected_person	main	5	Person_ID	1	0
affected_person_emergency_item	relationship (link)	2	Person_ID, Item_ID	0	0
area	main	6	Area_ID	2	0
damage	main	7	Damage_ID	2	2
disaster	main	5	Disaster_ID	0	0
disaster_affected_person	relationship (link)	2	Disaster_ID, Person_ID	0	0
disaster_area	relationship (link)	3	Disaster_ID, Area_ID	0	0
disaster_rescue_org	relationship (link)	2	Disaster_ID, Org_ID	0	0
disaster_shelter	relationship (link)	2	Disaster_ID, Shelter_ID	0	0
early_warning	main	6	Alert_ID	1	0
emergency_item	main	3	Item_ID	0	0
relief_item	main	4	Relief_ID	1	0
rescue_org	main	4	Org_ID	0	0
rescue_org_relief_item	relationship (link)	2	Org_ID, Relief_ID	0	0
shelter	main	7	Shelter_ID	3	0
shelter_emergency_item	relationship (link)	2	Shelter_ID, Item_ID	0	0
shelter_assignment	main	6	Assignment_ID	1	2
relief_distribution	main	7	Distribution_ID	1	2
relief_recipient	relationship (link)	2	Distribution_ID, Person_ID	0	2
disaster_track	main	8	Track_ID	4	0
users	stand-alone	7	User_ID	0	0
audit_log	stand-alone	7	Log_ID	0	0

Unique keys (8)

Table	Key	Columns
area	uq_area_name_location	Area_Name, Location
emergency_item	uq_emergency_item_name	Item_Name
relief_item	uq_relief_item_name	Item_Name
rescue_org	uq_rescue_org_name	Org_Name
shelter	uq_shelter_name	Shelter_Name
shelter_assignment	uq_assignment_disaster_person	Disaster_ID, Person_ID
disaster_track	uq_track_disaster_time	Disaster_ID, Point_Time
users	uq_users_username	Username

ENUM columns (5)

Column	Values
area.Risk_Level	Low,Medium,High,Very High
disaster.Status	Active,Ended
disaster_area.Relief_Shortage	No,Yes
disaster_track.Point_Type	Actual,Forecast
users.Role	admin,editor,viewer

Triggers (8)

Trigger	Table	When
trg_assignment_ins	shelter_assignment	BEFORE INSERT
trg_assignment_upd	shelter_assignment	BEFORE UPDATE
trg_distribution_ins	relief_distribution	BEFORE INSERT
trg_distribution_upd	relief_distribution	BEFORE UPDATE
trg_recipient_ins	relief_recipient	BEFORE INSERT
trg_recipient_upd	relief_recipient	BEFORE UPDATE
trg_damage_ins	damage	BEFORE INSERT
trg_damage_upd	damage	BEFORE UPDATE

Triggers refuse a stay, distribution, recipient or damage record whose shelter, family, organisation or area does not belong to that disaster. Capacity and stock limits are checked by the application (inc/rules.php) inside a transaction with row locks.